

Sequential finetuning											
recognition	30.4	10.8	0.5	0.5	1.2	15.4	33.0	26.4	18.5	4.5	14.1
location	7.0	29.0	0.1	0.1	0.0	4.8	0.1	4.3	5.2	1.0	5.2
judge	0.5	0.8	63.9	61.5	0.2	41.0	0.1	0.1	23.2	7.0	19.8
commonsense	0.7	2.1	62.7	66.7	0.2	41.2	0.1	0.1	23.3	8.0	20.5
count	0.3	0.2	0.1	0.1	34.6	0.1	0.0	0.2	0.5	0.4	3.6
action	11.6	1.9	56.0	29.8	0.1	55.9	0.1	6.8	22.8	7.8	19.3
color	1.6	0.3	0.1	0.0	0.0	0.1	51.0	3.2	0.5	0.0	5.7
type	12.3	10.6	0.3	0.3	0.1	8.1	1.8	27.3	13.1	7.0	8.1
subcategory	9.7	17.7	55.3	61.6	0.8	43.3	5.1	16.2	45.2	6.9	26.2
causal	7.6	10.2	26.4	37.6	0.1	16.0	0.2	11.2	29.7	10.2	14.9
	q_recognition	q_location	q_judge	q_commonsense	q_count	q_action	q_color	q_type	q_subcategory	q_causal	Avg

$\mathcal{L}_{QR}$											
recognition	30.4	10.8	0.5	0.5	1.2	15.4	33.0	26.4	18.5	4.5	14.1
location	26.4	30.4	0.5	0.2	0.9	14.1	28.4	23.4	20.1	3.2	14.8
judge	21.6	29.1	64.7	1.6	1.1	53.7	23.4	18.7	18.4	3.9	23.6
commonsense	19.0	25.2	63.2	66.0	0.2	51.5	21.0	17.9	35.7	6.8	30.7
count	15.0	22.3	62.0	64.4	35.7	47.9	18.0	14.9	32.7	8.0	32.1
action	19.1	23.4	62.5	67.8	35.4	57.4	17.5	18.4	35.2	10.4	34.7
color	16.1	22.4	61.6	67.8	34.8	53.9	51.3	14.0	33.9	11.4	36.7
type	17.1	22.0	61.5	67.7	34.5	50.8	46.5	33.9	38.7	11.1	38.4
subcategory	18.4	24.2	61.6	66.3	32.8	52.4	45.6	26.2	48.9	10.4	38.7
causal	16.2	24.2	35.0	57.0	33.1	25.3	43.7	23.0	39.3	10.4	30.7
	recognition	location	judge	commonsense	count	action	color	type	subcategory	causal	Avg

QUAD (Ours)											
recognition	30.4	10.8	0.5	0.5	1.2	15.4	33.0	26.4	18.5	4.5	14.1
location	28.0	28.4	0.5	0.4	0.9	14.7	28.7	24.3	20.9	3.6	15.1
judge	24.0	28.9	63.9	1.0	0.9	54.4	24.8	20.5	19.2	2.8	24.0
commonsense	21.8	29.0	62.7	69.1	0.3	52.7	23.3	20.9	40.0	9.4	32.9
count	19.9	28.3	61.2	67.6	35.3	51.9	20.0	18.4	37.3	7.6	34.8
action	21.7	28.9	63.6	68.7	35.5	58.4	18.6	19.9	39.8	9.2	36.4
color	20.0	27.2	62.3	67.9	35.2	55.7	52.2	15.1	38.4	11.7	38.6
type	22.4	25.6	62.3	68.4	35.4	55.5	48.4	33.9	40.2	10.2	40.2
subcategory	22.1	26.0	62.6	67.4	35.3	54.5	46.2	29.4	49.6	9.7	40.3
causal	20.4	25.8	62.6	65.1	35.6	54.4	45.6	24.1	44.8	14.1	39.3
	recognition	location	judge	commonsense	count	action	color	type	subcategory	causal	Avg